Breast Cancer SVM Classification Report

Machine Learning Pipeline & Dataset Overview | Kaggle Execution Summary

Executive Summary

This document details the analysis and modeling pipeline for the Synthetic Breast Cancer dataset using Support Vector Machines (SVM). The goal is binary classification of tissue samples as either **Benign** or **Malignant** based on 30 standardized numerical features derived from cell nuclei imagery.

TOTAL OBSERVATIONS
569

TOTAL FEATURES
32

TARGET CLASS DISTRIBUTION
360 B / 209 M

MISSING VALUES
0 (0.0%)

Dataset Structure & Sample Record

The dataset consists of 569 records with 30 continuous floating-point metrics, one unique identifier (`sample_id`), and one categorical target variable (`diagnosis`). Below is a preview of the initial observations:

ID	Radius Mean	Texture Mean	Perimeter Mean	Area Mean	Smoothness Mean	Diagnosis
1	-5.6767	1.6414	1.6629	-1.3240	1.3246	Malignant
2	0.6105	1.1004	-0.9745	0.3421	0.9564	Benign
3	-6.4476	-1.3248	0.7572	-0.0325	-1.0825	Benign
4	0.6192	0.4459	1.3351	-0.1726	-4.4752	Benign
5	2.3300	0.2315	0.7907	0.8324	-1.5858	Benign

Target Class Balance Analysis

The dataset shows a moderate class imbalance favoring benign diagnoses, reflecting typical medical diagnostic distributions where non-cancerous findings are more frequent.



Classification Pipeline Implementation

The notebook utilizes Scikit-Learn to construct a robust classification workflow. Below is the structured Python setup used to load the dataset dynamically and prepare the modeling environment:

```
import glob import pandas as pd from sklearn.model_selection import train_test_split from sklearn.pipeline import Pipeline from sklearn.preprocessing import StandardScaler from sklearn.svm import SVC # Dynamic file lookup on Kaggle environment file_list = glob.glob("/kaggle/input/**/synthetic_breast_cancer_svm_dataset.csv", recursive=True) df = pd.read_csv(file_list[0]) # Preprocessing & Model Building Pipeline pipeline = Pipeline([ ('scaler', StandardScaler()), ('svm', SVC(kernel='rbf', C=1.0)) ])
```

Data Quality & Preprocessing Summary

Completeness: All 32 columns contain 569 non-null entries, eliminating the need for missing-value imputation.
Feature Scaling Requirement: Features are synthetic standardized metrics with varied scales. Feature scaling via `StandardScaler` is required prior to applying Support Vector Classifiers (SVC) to ensure optimal hyper-plane margin calculation.